

Lilja Kiiski

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EDUCATION

Purdue University, West Lafayette, IN

B.S in Electrical Engineering, Honors | Minor in Mathematics

Completed: EE Fundamentals I, Engineering Systems Design, Differential Equations, Linear Algebra

In Progress: EE Fundamentals II, Digital Design, Advanced C Programming, E&M, Microprocessors

May 2029

GPA: 3.81/4.00

TECHNICAL EXPERIENCE

CS Extended Reality (XR) Lab — Undergraduate Researcher

Aug 2026 - Present

West Lafayette, IN

- Designing a haptic device concept using three orthogonal voice coil actuators, with an ESP32 and IMU to render 3D force-direction feedback, simulating electric field forces for extended reality applications
- Planned a simple hand-driven prototype with faculty mentor to test whether users can perceptually distinguish force direction, before building the full device

Purdue University College of Engineering, Honors — Undergraduate Teaching Assistant

Aug 2026 - Present

West Lafayette, IN

- Supported 50+ students with in-class activities, labs, projects, and homework focused on MATLAB and Python programming, physics, and fundamental engineering design concepts
- Created and taught custom review sessions to aid student preparation for midterms and exams

Purdue IEEE Autonomous Motorsports — Electrical Team Member

Aug 2025 - May 2026

West Lafayette, IN

- Designed the schematic for an 84V-to-19V/12V DC-DC converter PCB powering team's autonomous karts lidar system, working from system requirements and engineering drawings
- Simulated core circuit components in LTSpice under mentor guidance prior to physical layout and fabrication

Monta Vista Robotics Team — President, Computer Vision Lead

Oct 2021 - Jun 2025

Cupertino, CAA

- Directed 150 member team through full design, build, and test development of competitive FRC robot
- Led software division in developing team's first AprilTag localization and alignment vision system
- Managed team schedule, budget, and communication with team mentors and sponsors

TECHNICAL PROJECTS

4-Bit Carry-Save Adder

Jan - May 2026

- Collaborated within a team of 4 to design and build a physical 4-bit Carry-Save Adder summing three 4-bit numbers
- Developed a SystemVerilog testbench to conduct functional testing and report results across all 4,096 input combinations, verifying project functionality

AprilTag Computer Vision System

Oct 2022 - Apr 2025

- Built a Linux-based AprilTag detection pipeline enabling real-time robot self-localization on the game field
- Implemented alignment logic directing the robot to the nearest scoring location for game-piece deployment

Honeycomb Game Console

May - Jun 2025

- Engineered a miniature embedded game console with button/joystick input and a TFT LCD display
- Designed a custom honeycomb-inspired enclosure in Onshape, fabricated via 3D printing

SKILLS

Software & Tools: C, C++, Java, Python, MATLAB, Linux, Bash/Shell, Git, SSH, Raspberry Pi

Design & Simulation Tools: SystemVerilog, KiCad, LTSpice, Onshape, Arduino, 3D Printing

Lab Equipment & Fabrication: Oscilloscope, Digital Multimeter, Breadboarding, Soldering, Crimping, 3D Printing

Languages: English (Native), Finnish (Native), Spanish (Elementary)